

Fig. 1: Object counter circuit

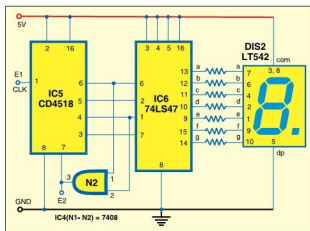


Fig. 2: Expansion circuit

in Fig. 1 and the expansion circuit in Fig. 2. For instance, to count the ob-

ject's IR transmitter (IR LED1) is placed on one side of the conveyor belt on which objects move. On the other side is a phototransistor L14F1 (T1) that senses the transmitted IR beam. When an object passes between the transmitter and receiver, it obstructs the IR beam. This pulls the base of npn transistor BC547 (T2) high, driving the transistor into saturation and pulling the trigger pin (pin 2)

of NE555 timer low.

The NE555 timer (IC1) is configured as a monostable multivibrator that generates a pulse of 330ms duration. This monostable output is given as clock to dual up-counter CD4518 (IC2). IC2 increments on every positive edge of the clock, giving a BCD output on pins 3 to 6. The purpose of NE555 timer is to prevent

signal debouncing from transistor T2. NE555 timer gives a perfect clock pulse.

When the BCD count reaches 9, IC2 is reset by AND gate N1 (IC4), simultaneously sending a clock signal to counter CD4518 (IC5) of the expansion circuit (if used). This BCD value is given to seven-segment display driver 74LS47 (IC3), which displays the number on seven-segment display DIS1. The count can be easily increased by connecting E1 pin from IC4 in Fig. 1 to E1 pin of the expansion circuit in Fig. 2. The E2 pin in Fig. 2 can be used to clock another similar expansion circuit. This way the objects' count can be increased as desired by simply adding an expansion circuit. ●

Solar Lamp with Variable Power Supply and Charger

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A simple circuit of a solar lamp with variable power supply and charger is described here. It is useful where mains power failure is frequent. The power supply can be used for testing electronics projects and also charging your mobile phone battery. You can save on your electricity bills by switching to an alternative source of power using this circuit.

The circuit works with the solar panel (without mains power supply) and can be used as a solar lamp during daytime (with the solar panel), night lamp (with external 12V battery), variable power supply (1.5V to 15V) and mobile battery (3.6V) charger.

The photovoltaic module or solar panel used here is capable of delivering a power of 5 watts. At full sunlight, it outputs 16.5V. The solar panel can deliver a current of 300-350 mA. Using

it you can charge Li-ion batteries common in mobile phones.

The working of the circuit is simple. The output of the solar panel is fed to the circuit via diode 1N4007 (D3), which acts as a polarity guard to protect the solar panel.

For using the small LED-based lamp in daytime, just flip switch S1 to

